

MH3100 Real Analysis I

Tutorial 1 Solutions

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Overview

- **Lecture coverage:** Lectures 1–3, namely the construction of $\mathbb{R}$, the Axiom of Completeness, suprema, infima, maxima, minima, the Nested Interval Property, and the Archimedean Property.
- **Questions:** 8 questions in total, namely Q1 through Q5 and Additional Questions A1 through A3.
- **Notation:** $\mathbb{N} = \{1, 2, 3, \dots\}$, and 0 is *not* a natural number in this course; sup, inf, max, min are written with the built-in operators; ε denotes ε .

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Question 1 (Suprema, infima, maxima, minima of four sets)

Problem

Compute, without proofs, the suprema and infima (if exist) of the following sets, and determine if the suprema and infima (if exist) are, respectively, maxima and minima of the respective sets.

(a) $A = \left\{ \frac{m}{n} : m, n \in \mathbb{N} \text{ with } m \leq n \right\}.$

(b) $B = \left\{ \frac{(-1)^m}{n} : m, n \in \mathbb{N} \right\}.$

(c) $C = \left\{ \frac{n}{3n+1} : n \in \mathbb{N} \right\}.$

(d) $D = \left\{ \frac{m}{m+n} : m, n \in \mathbb{N} \right\}.$

(Tutorial 1, Q1.)

Solution

Part (a)

Proof (supremum). Let $m, n \in \mathbb{N}$ with $m \leq n$.

$$\frac{m}{n} \leq \frac{n}{n} = 1$$

$\therefore 1$ is an upper bound for A .

$$1 = \frac{1}{1} \in A$$

$\therefore 1 = \max A$, and $\max A = \sup A$. □

$\sup A = \max A = 1.$

Proof (infimum). Let $m, n \in \mathbb{N}$.

$$\frac{m}{n} \geq \frac{1}{n} > 0$$

$\therefore 0$ is a lower bound for A , and $0 \notin A$. Let $\varepsilon > 0$; choose $n \in \mathbb{N}$ with $n > \frac{1}{\varepsilon}$ (Archimedean).

$$\frac{1}{n} \in A \quad (m = 1 \leq n), \quad \frac{1}{n} < 0 + \varepsilon$$

$\therefore 0 = \inf A$ (ε -Criterion). $0 \notin A \therefore 0$ is not a minimum. □

$\inf A = 0, \quad 0 \text{ is not the minimum of } A.$

Part (b)**Proof (supremum).** Let $m, n \in \mathbb{N}$.

$$\frac{(-1)^m}{n} \leq \frac{1}{n} \leq 1$$

 $\therefore 1$ is an upper bound for B .

$$1 = \frac{(-1)^2}{1} \in B$$

 $\therefore 1 = \max B = \sup B$. □

$$\boxed{\sup B = \max B = 1.}$$

Proof (infimum). Let $m, n \in \mathbb{N}$.

$$\frac{(-1)^m}{n} \geq -\frac{1}{n} \geq -1$$

 $\therefore -1$ is a lower bound for B .

$$-1 = \frac{(-1)^1}{1} \in B$$

 $\therefore -1 = \min B = \inf B$. □

$$\boxed{\inf B = \min B = -1.}$$

Part (c)**Proof (infimum).** Let $c_n := \frac{n}{3n+1}$. For $n \in \mathbb{N}$:

$$\begin{aligned} c_{n+1} - c_n &= \frac{n+1}{3n+4} - \frac{n}{3n+1} \\ &= \frac{(n+1)(3n+1) - n(3n+4)}{(3n+4)(3n+1)} \\ &= \frac{1}{(3n+4)(3n+1)} > 0 \end{aligned}$$

 $\therefore c_1 < c_2 < c_3 < \dots$ (induction). $\therefore c_n \geq c_1 = \frac{1}{4}$ for all $n \in \mathbb{N}$, and $c_1 \in C$. $\therefore \frac{1}{4} = \min C = \inf C$. □

$$\boxed{\inf C = \min C = \frac{1}{4}.}$$

Proof (supremum). For $n \in \mathbb{N}$:

$$\frac{1}{3} - c_n = \frac{1}{3(3n+1)} > 0$$

$\therefore \frac{1}{3}$ is an upper bound for C , and $\frac{1}{3} \notin C$. Let $\varepsilon > 0$; choose $n \in \mathbb{N}$ with $n > \frac{1}{\varepsilon}$ (Archimedean).

$$\begin{aligned} \frac{1}{3} - c_n &= \frac{1}{3(3n+1)} \\ &< \frac{1}{n} & (3(3n+1) > n) \\ &< \varepsilon \end{aligned}$$

$\therefore c_n \in C$ and $c_n > \frac{1}{3} - \varepsilon$. $\therefore \frac{1}{3} = \sup C$ (ε -Criterion). $\frac{1}{3} \notin C \therefore \frac{1}{3}$ is not a maximum. $\square$

$\sup C = \frac{1}{3}, \quad \frac{1}{3} \text{ is not the maximum of } C.$

Part (d)

Proof (supremum). Let $m, n \in \mathbb{N}$.

$$m < m + n \implies \frac{m}{m + n} < 1$$

$\therefore 1$ is an upper bound for D , and $1 \notin D$. Let $\varepsilon > 0$; choose $m \in \mathbb{N}$ with $m > \frac{1}{\varepsilon}$ (Archimedean). Then $\frac{m}{m+1} \in D$ ($n = 1$).

$$\begin{aligned} 1 - \frac{m}{m+1} &= \frac{1}{m+1} \\ &< \frac{1}{m} < \varepsilon \end{aligned}$$

$\therefore \frac{m}{m+1} \in D$ and $\frac{m}{m+1} > 1 - \varepsilon$. $\therefore 1 = \sup D$ (ε -Criterion). $1 \notin D \therefore 1$ is not a maximum. $\square$

$\sup D = 1, \quad 1 \text{ is not the maximum of } D.$

Proof (infimum). Let $m, n \in \mathbb{N}$.

$$\frac{m}{m+n} > 0$$

$\therefore 0$ is a lower bound for D , and $0 \notin D$. Let $\varepsilon > 0$; choose $n \in \mathbb{N}$ with $n > \frac{1}{\varepsilon}$ (Archimedean). Then $\frac{1}{1+n} \in D$ ($m = 1$).

$$\begin{aligned} \frac{1}{1+n} &< \frac{1}{n} \\ &< \varepsilon = 0 + \varepsilon \end{aligned}$$

$\therefore 0 = \inf D$ (ε -Criterion). $0 \notin D \therefore 0$ is not a minimum. $\square$

$\inf D = 0, \quad 0 \text{ is not the minimum of } D.$

Question 2 (Supremum of a union: $\max\{\sup A, \sup B\} = \sup(A \cup B)$)

Problem

Prove that if the sets $A, B \subseteq \mathbb{R}$ are nonempty and bounded above, then

$$\max\{\sup A, \sup B\} = \sup(A \cup B).$$

(Tutorial 1, Q2; this is also the sample midterm Q3.)

Solution

Method 1: definition of supremum

Proof. $s := \sup A$ and $t := \sup B$ exist (AoC). Let $M := \max\{s, t\}$ and $x \in A \cup B$.

$$x \in A \implies x \leq s \leq M$$

$$x \in B \implies x \leq t \leq M$$

$\therefore M$ is an upper bound for $A \cup B \neq \emptyset$. $\therefore u := \sup(A \cup B)$ exists (AoC), and $u \leq M$ (least upper bound). $A \subseteq A \cup B \therefore u$ is an upper bound for $A \therefore s \leq u$ (def. of supremum). Likewise $t \leq u \therefore M \leq u$. $\therefore u \leq M \leq u$, i.e. $\sup(A \cup B) = M$. $\square$

$$\boxed{\max\{\sup A, \sup B\} = \sup(A \cup B).}$$

Method 2: ε -Criterion

Proof. Let $s := \sup A$, $t := \sup B$, $M := \max\{s, t\}$; M is an upper bound for $A \cup B$ (Method 1). WLOG $M = s$. Let $\varepsilon > 0$. There is $a \in A$ with $a > s - \varepsilon = M - \varepsilon$ (ε -Criterion for A), and $a \in A \cup B$. $\therefore M = \sup(A \cup B)$ (ε -Criterion). $\square$

$$\boxed{\max\{\sup A, \sup B\} = \sup(A \cup B).}$$

Question 3 ($\sup B = \inf A$ for the set of lower bounds)**Problem**

Suppose that the set $A \subseteq \mathbb{R}$ is nonempty and bounded below, and define the set

$$B := \{b \in \mathbb{R} : b \text{ is a lower bound for } A\}.$$

Prove that $\sup B = \inf A$. (*Tutorial 1, Q3.*)

Solution**Method 1: verification against the definition of infimum**

Proof. A is bounded below $\therefore B \neq \emptyset$. Fix $a_0 \in A$. For all $b \in B$: $b \leq a_0$ (b is a lower bound for A). $\therefore B$ is nonempty and bounded above $\therefore \sup B$ exists (AoC).

Lower bound. Let $a \in A$. For all $b \in B$: $b \leq a$. $\therefore a$ is an upper bound for B . $\therefore \sup B \leq a$. $\therefore \sup B$ is a lower bound for A .

Greatest. Let c be a lower bound for A . $\therefore c \in B \therefore c \leq \sup B$.

$\therefore \sup B$ is the greatest lower bound for A , i.e. $\sup B = \inf A$ (def. of infimum). $\square$

$\sup B = \inf A.$

Question 4 (Supremum of a sumset: $\sup(A+B) = \sup A + \sup B$)

Problem

Suppose that the sets $A, B \subseteq \mathbb{R}$ are nonempty and bounded above and that $s = \sup A$ and $t = \sup B$. Define the notation

$$A + B := \{a + b : a \in A \text{ and } b \in B\}.$$

- (a) Prove that $s + t$ is an upper bound for $A + B$.
- (b) Prove that if $u \in \mathbb{R}$ is an upper bound for $A + B$ and $a \in A$, then

$$t \leq u - a.$$

- (c) Hence deduce that $\sup(A + B) = s + t$.

(Tutorial 1, Q4.)

Solution

Method 1: upper-bound chase

(a).

Proof. Let $c \in A + B$, so $c = a + b$ with $a \in A$, $b \in B$.

$$a \leq s, \quad b \leq t \implies c = a + b \leq s + t$$

$\therefore s + t$ is an upper bound for $A + B$. □

$s + t$ is an upper bound for $A + B$.

(b).

Proof. Let u be an upper bound for $A + B$, and $a \in A$. For all $b \in B$:

$$a + b \in A + B \implies a + b \leq u \implies b \leq u - a$$

$\therefore u - a$ is an upper bound for B . $\therefore t \leq u - a$ (def. of supremum). □

$t \leq u - a$.

(c).

Proof. Let u be an upper bound for $A + B$. For all $a \in A$:

$$t \leq u - a \text{ (b)} \implies a \leq u - t$$

$\therefore u - t$ is an upper bound for A . $\therefore s \leq u - t$, i.e. $s + t \leq u$. With (a): $s + t$ is the least upper bound for $A + B$. $\therefore \sup(A + B) = s + t$ (def. of supremum). □

$\sup(A + B) = s + t$.

Method 2: ε -Criterion (part (c))

$s + t$ is an upper bound for $A + B$ (a). Let $\varepsilon > 0$. There are $a \in A$, $b \in B$ with $a > s - \frac{\varepsilon}{2}$ and $b > t - \frac{\varepsilon}{2}$ (ε -Criterion).

$$a + b \in A + B, \quad a + b > s + t - \varepsilon$$

$\therefore \sup(A + B) = s + t$ (ε -Criterion).

Question 5 (An ε -characterisation of the supremum)

Problem

Prove that if $A \subseteq \mathbb{R}$ is nonempty and bounded above, and $s \in \mathbb{R}$ has the property that

$$\forall n \in \mathbb{N} \quad s + \frac{1}{n} \text{ is an upper bound for } A \text{ and } s - \frac{1}{n} \text{ is not an upper bound for } A,$$

then $s = \sup A$. (*Tutorial 1, Q5.*)

Solution

Method 1: contradiction via the Archimedean Property

Proof. $s_0 := \sup A$ exists (AoC). Suppose $s \neq s_0$.

Case $s < s_0$. Choose $n \in \mathbb{N}$ with $\frac{1}{n} < s_0 - s$ (Archimedean).

$$s + \frac{1}{n} < s_0$$

$s + \frac{1}{n}$ is an upper bound for A (hypothesis) $\therefore s_0 \leq s + \frac{1}{n}$ (def. of supremum). This contradicts $s + \frac{1}{n} < s_0$.

Case $s > s_0$. Choose $n \in \mathbb{N}$ with $\frac{1}{n} < s - s_0$ (Archimedean).

$$\forall a \in A : \quad a \leq s_0 < s - \frac{1}{n}$$

$\therefore s - \frac{1}{n}$ is an upper bound for A . This contradicts the hypothesis.

$\therefore s = s_0 = \sup A$. □

$s = \sup A.$

Question A1 (Supremum of a countable union)

Problem

Suppose that $A_1, A_2, A_3, \dots$ is a list of nonempty sets, each of which is bounded above. Find a formula for $\sup(\bigcup_{n=1}^{\infty} A_n)$ in terms of $\sup A_1, \sup A_2, \sup A_3, \dots$ (*Tutorial 1, Additional Question A1.*)

Solution

Method 1: two inequalities, case split on boundedness

Proof. Let $U := \bigcup_{n=1}^{\infty} A_n \neq \emptyset$ and $S := \{\sup A_n : n \in \mathbb{N}\}$. $\sup A_n$ exists for each $n \in \mathbb{N}$ (AoC) $\therefore S \neq \emptyset$.

Case 1: S bounded above. $M := \sup S$ exists (AoC). Let $x \in U$, so $x \in A_n$ for some n .

$$x \leq \sup A_n \leq M$$

$\therefore M$ is an upper bound for U . $\therefore \sup U$ exists (AoC) and $\sup U \leq M$. For each $n \in \mathbb{N}$: $A_n \subseteq U \therefore \sup U$ is an upper bound for $A_n \therefore \sup A_n \leq \sup U$. $\therefore \sup U$ is an upper bound for $S \therefore M \leq \sup U$. $\therefore \sup U = M = \sup S$.

Case 2: S not bounded above. Let $K \in \mathbb{R}$. Some n has $\sup A_n > K$. $\therefore K$ is not an upper bound for A_n (else $\sup A_n \leq K$). $\therefore x > K$ for some $x \in A_n \subseteq U$. $\therefore U$ is not bounded above, and $\sup U = \infty = \sup S$.

$\therefore \sup U = \sup S$ in both cases. □

$$\sup \left(\bigcup_{n=1}^{\infty} A_n \right) = \sup \{ \sup A_n : n \in \mathbb{N} \}.$$

Question A2 (True/false: nested and non-nested interval intersections)

Problem

Determine if each of the following statements is true.

- (a) There is a sequence of nested open intervals $J_1 \supseteq J_2 \supseteq J_3 \supseteq \cdots$ with $\bigcap_{n=1}^{\infty} J_n$ nonempty but containing only a finite number of elements.
- (b) There is a sequence of closed and bounded (but not necessarily nested) intervals $I_1, I_2, I_3, \dots$ with the property that $\bigcap_{n=1}^N I_n \neq \emptyset$ for all $N \in \mathbb{N}$, but $\bigcap_{n=1}^{\infty} I_n = \emptyset$.

(Tutorial 1, Additional Question A2.)

Solution

Part (a): True

Proof. Let $J_n := \left(-\frac{1}{n}, \frac{1}{n}\right)$, an open interval for each $n \in \mathbb{N}$.

$$\frac{1}{n+1} < \frac{1}{n} \implies J_{n+1} \subseteq J_n$$

$\therefore J_1 \supseteq J_2 \supseteq \cdots$ is nested. $0 \in J_n$ for all $n \therefore 0 \in \bigcap_{n=1}^{\infty} J_n$. Let $x \neq 0$; choose $n \in \mathbb{N}$ with $\frac{1}{n} < |x|$ (Archimedean). $\therefore x \notin J_n \therefore x \notin \bigcap_{n=1}^{\infty} J_n \therefore \bigcap_{n=1}^{\infty} J_n = \{0\}$, nonempty with exactly one element. $\square$

Statement (a) is TRUE; witnessed by $J_n = (-1/n, 1/n)$, $\bigcap_{n=1}^{\infty} J_n = \{0\}$.

Part (b): False

Proof. Let $I_n = [a_n, b_n]$ with $K_N := \bigcap_{n=1}^N I_n \neq \emptyset$ for all $N \in \mathbb{N}$. We show $\bigcap_{n=1}^{\infty} I_n \neq \emptyset$, so no such sequence exists. Put $\alpha_N := \max\{a_1, \dots, a_N\}$ and $\beta_N := \min\{b_1, \dots, b_N\}$.

$$x \in K_N \iff a_n \leq x \leq b_n \ (n = 1, \dots, N) \iff \alpha_N \leq x \leq \beta_N$$

$\therefore K_N = [\alpha_N, \beta_N]$ is a nonempty closed bounded interval.

$$K_{N+1} = K_N \cap I_{N+1} \subseteq K_N$$

$\therefore K_1 \supseteq K_2 \supseteq \cdots$ is nested. $\therefore \bigcap_{N=1}^{\infty} K_N \neq \emptyset$ (Nested Interval Property). $K_N \subseteq I_n$ for $N \geq n \therefore \bigcap_{N=1}^{\infty} K_N \subseteq \bigcap_{n=1}^{\infty} I_n \therefore \bigcap_{n=1}^{\infty} I_n \neq \emptyset$. $\square$

Statement (b) is FALSE (via the Nested Interval Property applied to K_N .)

Question A3 (The set $\mathbb{N} \times \mathbb{N}$ is countable)

Problem

Prove that $\mathbb{N} \times \mathbb{N}$ is countable; i.e., elements of $\mathbb{N} \times \mathbb{N}$ can be listed in a sequence indexed by the natural numbers. (*Tutorial 1, Additional Question A3.*)

Solution

Method 1: diagonal enumeration

Proof. Let $T_k := \frac{k(k+1)}{2}$, so $T_0 = 0$.

$$T_k - T_{k-1} = k \implies 0 = T_0 < T_1 < T_2 < \dots, \quad T_k \geq k$$

For $p \in \mathbb{N}$ let k be least with $T_k \geq p$ (exists, since $T_p \geq p$); then $T_{k-1} < p \leq T_k$. $\therefore \mathbb{N}$ is the disjoint union of the blocks $(T_{k-1}, T_k]$, $k \in \mathbb{N}$. Define $f(m, n) := \frac{(m+n-2)(m+n-1)}{2} + m$, and put $k := m+n-1 \in \mathbb{N}$.

$$1 \leq m \leq k, \quad f(m, n) = T_{k-1} + m \in (T_{k-1}, T_k]$$

$\therefore f : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ is well defined.

Injective. Let $(m, n), (m', n') \in \mathbb{N} \times \mathbb{N}$ with $f(m, n) = f(m', n')$; put $k := m+n-1$, $k' := m'+n'-1$. $\therefore$ the common value lies in $(T_{k-1}, T_k]$ and in $(T_{k'-1}, T_{k'}]$, which are disjoint unless $k = k'$, so $k = k'$. $\therefore T_{k-1} + m = T_{k-1} + m'$, so $m = m'$ and $n = k+1-m = n'$.

Surjective. Let $p \in \mathbb{N}$ with $T_{k-1} < p \leq T_k$. Put $m := p - T_{k-1} \in \{1, \dots, k\}$ and $n := k+1-m \in \mathbb{N}$.

$$m+n-1 = k \implies f(m, n) = T_{k-1} + m = p$$

$\therefore f$ is a bijection onto $\mathbb{N}$. $\therefore$ listing (m, n) in the order $f(m, n) = 1, 2, 3, \dots$ lists $\mathbb{N} \times \mathbb{N}$ in a sequence indexed by $\mathbb{N}$. $\square$

$\mathbb{N} \times \mathbb{N} \text{ is countable, via the bijection } f(m, n) = \frac{(m+n-2)(m+n-1)}{2} + m.$

Method 2: injection $(m, n) \mapsto 2^m 3^n$

Proof. Let $g(m, n) := 2^m 3^n$ and $(m, n), (m', n') \in \mathbb{N} \times \mathbb{N}$ with $g(m, n) = g(m', n')$. If $m > m'$: $2^{m-m'} 3^n = 3^{n'}$ has an even left side and an odd right side, a contradiction; likewise $m < m'$. $\therefore m = m'$, so $3^n = 3^{n'}$. $\therefore n = n'$ ($t \mapsto 3^t$ strictly increasing). $\therefore g$ is injective. Put $x_k := g^{-1}(k)$ if $k \in g(\mathbb{N} \times \mathbb{N})$, and $x_k := (1, 1)$ otherwise.

$$(m, n) = x_{g(m, n)} \quad \text{for all } (m, n) \in \mathbb{N} \times \mathbb{N}$$

$\therefore x_1, x_2, x_3, \dots$ lists every element of $\mathbb{N} \times \mathbb{N}$. Deleting each repeat of an earlier entry leaves a list (infinite) with each pair exactly once. $\square$

Method Map

Q.	Core topic	Methods
1	Sup/inf/max/min of explicit sets	Bound + attaining element; ε -Criterion with Archimedean Property.
2	$\max\{\sup A, \sup B\} = \sup(A \cup B)$	Definition of supremum; ε -Criterion (Method 2).
3	$\sup B = \inf A$ for the lower-bound set	Verification against the definition of infimum.
4	$\sup(A + B) = \sup A + \sup B$	Upper-bound chase (a)–(c); ε -Criterion with $\varepsilon/2$ (Method 2).
5	ε -characterisation of $\sup A$	Contradiction by cases, Archimedean Property.
A1	sup of a countable union	Two inequalities; case split on boundedness.
A2	Interval intersections	Explicit example (a); running intersections and Nested Interval Property (b).
A3	Countability of $\mathbb{N} \times \mathbb{N}$	Diagonal bijection via triangular numbers; injection $2^m 3^n$ (Method 2).

Final Answers

Q1(a) $\sup A = \max A = 1$; $\inf A = 0$, not the minimum.

Q1(b) $\sup B = \max B = 1$; $\inf B = \min B = -1$.

Q1(c) $\sup C = \frac{1}{3}$, not the maximum; $\inf C = \min C = \frac{1}{4}$.

Q1(d) $\sup D = 1$, not the maximum; $\inf D = 0$, not the minimum.

Q2 $\max\{\sup A, \sup B\} = \sup(A \cup B)$.

Q3 $\sup B = \inf A$.

Q4 (a) $s + t$ is an upper bound for $A + B$. (b) $t \leq u - a$. (c) $\sup(A + B) = s + t$.

Q5 $s = \sup A$.

A1 $\sup(\bigcup_{n=1}^{\infty} A_n) = \sup\{\sup A_n : n \in \mathbb{N}\}$.

A2 (a) True: $J_n = (-1/n, 1/n)$, $\bigcap_{n=1}^{\infty} J_n = \{0\}$. (b) False.

A3 $\mathbb{N} \times \mathbb{N}$ is countable: $f(m, n) = \frac{(m+n-2)(m+n-1)}{2} + m$ is a bijection onto $\mathbb{N}$.